IRELE

Identification of dynamic systems

Silverbox System Identification

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1 Introduction

The identification of dynamical systems is a crucial topic for bridging the gap between real-world systems and their mathematical models. It provides valuable insights into the underlying dynamic behavior, enabling a wide range of applications such as control design, prediction, and fault diagnosis. Within this framework, an unknown system is provided with the sole purpose of applying identification tools studied during the course to extract the hidden system model from measured data. The system under study, referred to as SilverBox, is an electronic circuit designed to mimic the behavior of a second-order nonlinear mechanical system, namely a mass-spring-damper system. This report is organized into two main parts. The first part focuses on the identification of linear and nonlinear contributions using nonparametric identification methods. The second part is dedicated to estimating the underlying mathematical model and analyzing the accuracy and validity of the obtained estimation. Throughout this work, we adopt a subjective investigative approach, emphasizing the reasoning and decisions made during the identification process and, therefore, a first-person tone is used throughout the report.

2

Nonparametric identification

The objective of this chapter is to identify the underlying Linear Time-Invariant (**LTI**) system by extracting the linear, nonlinear, and noisy contributions from the output using two nonparametric approaches, namely the **Fast method** and the **Robust method**.

2.1 System and measurement devices

The system made available to us was shared without any prior information except its descriptive name, **SilverBox circuit**. this class of circuits is characterized by a second-order linear time-invariant system with a third order static non-linear feedback, which can be expressed by the equations 2.1 and 2.2.

$$m\frac{d^2 y(t)}{dt^2} + d\frac{dy(t)}{dt} + k\big(y(t)\big)\, y(t) = u(t) \tag{2.1}$$

$$k(y) = a + b y^2 \tag{2.2}$$

Nevertheless, it is reasonable to assume that the nonlinearity can be tuned based on annotations on an actuator (V^3, $|V|V^2$, V^2,..). For the time being, the actuator will be set to the position V^3.

The measurement generation and acquisition device, provided by the department, is referred to as *PXI measurement system*. It can generate multisine signals and is capable of acquiring measurements at sampling rate above 10 GHz.

2.2 Multisine excitation

2.2.1 Excitation class

The objective of estimating the underlying LTI system can be expressed as finding the **Best linear approximation** of the system from the output measurements, or in other words, as finding the model which minimizes the mean squared error between the modelled and measured output in steady-state. Mathematically, it boils down to the equation 2.3.

$$G_{\mathrm{BLA}} \equiv \arg\min_{G} \mathbb{E}\Big[|G_{\mathrm{NL}}(u) - G(u)||^2\Big] \tag{2.3}$$

From the course of *Identification of dynamical systems*, we know that, to evaluate the BLA, the system must be a **PISPO** system, meaning that a periodic input produces a periodic output with the same fundamental period. Moreover, it depends on the class of the excitation signal. Consequently, the amplitude of the nonlinear contributions varies with both the signal power and the distribution of power across the input frequency spectrum. This can be formalized by expressing the BLA as the sum of the true underlying system, $G_0(j\omega_k)$, and a bias contribution, $G_B(j\omega_k)$, due to the nonlinear part of the system which generates output components correlated with the input spectrum. These terms are referred to as **coherent** spectral components, $Y_{BLA}(k)$, in contrast to the uncorrelated terms (not independent as generated by the input) named as **incoherent** spectral components, $Y_s(k)$.

$$Y(k) = \big(G_0(j\omega_k) + G_B(j\omega_k)\big)\,U(k) + Y_s(k) \tag{2.4}$$

For the estimation and modeling of the underlying LTI system, the choice of excitation is crucial. It must remain consistent across all experiments to ensure a coherent analysis and allow for meaningful comparisons. Consequently, the BLA depends on the **class** of the excitation. Two excitations belong to the same class if they share the same *Riemann equivalent power spectrum*, meaning they have both the same signal power (identical root mean squared (RMS) amplitude) and the same power spectral density (PSD) distribution. In this regard, multisine excitations are particularly convenient, as their spectral power distribution can be easily tuned (each sine amplitude can be adjusted individually) and they are compatible with the two nonparametric identification approaches used in the following sections. Therefore, the estimated BLA is valid for excitations belonging to this class.

2.3 Input signal amplitude and band of interest

Considering the measurement devices at our disposal, it is first necessary to ensure that the input and output signals remain within the amplitude limits of the devices. This prevents clipping or saturation, which would result in distorted measurements and corrupted harmonics. Accordingly, the RMS amplitude of the input signal is fixed at 0.1 V (in *Matlab*, with a power multiplication factor of 6 on the PXI system), utilizing nearly the full ADC range, thereby minimizing quantization errors while avoiding saturation. These considerations are illustrated in Figures 2.2 and 2.1 for experiments using multisine excitations up to 8 kHz.

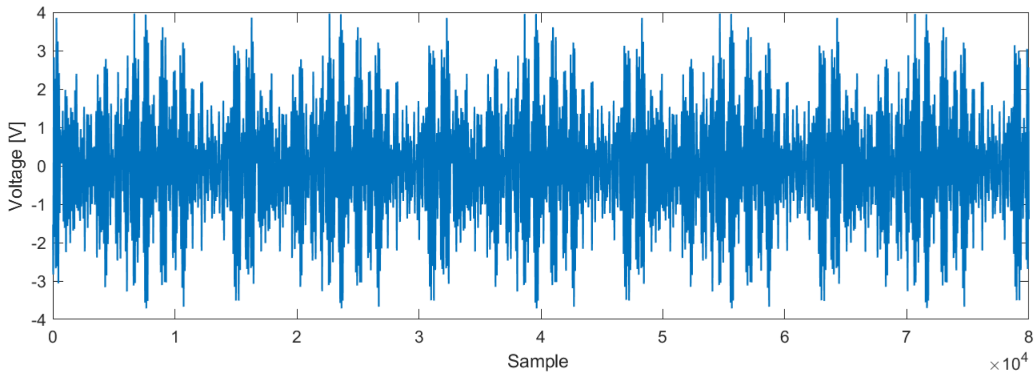


Figure 2.1: Output measurement - time domain - multisine up to 8 KHz

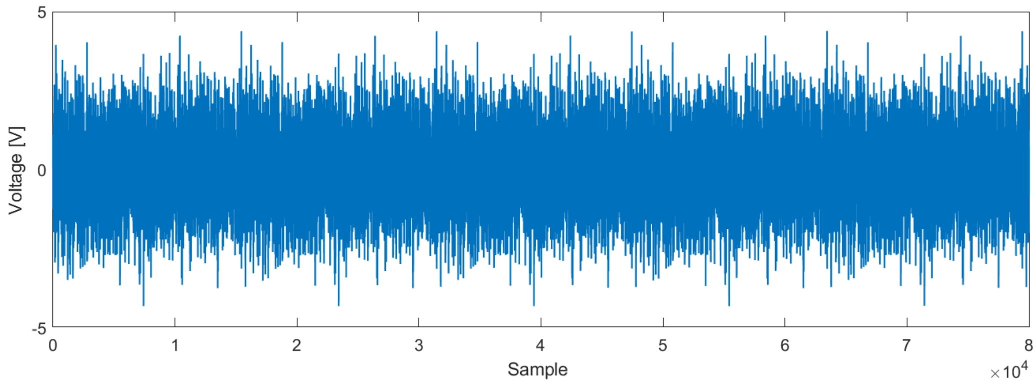


Figure 2.2: Input measurement - time domain - multisine up to 8 KHz

The figures show that the input signal is constrained under 5.5V, the PXI voltage limit.

Finally, since the system under evaluation is unknown, it is good practice to first excite the system with a very large signal bandwidth (multisine excitation up to 30 kHz) in order to find the band of interest.

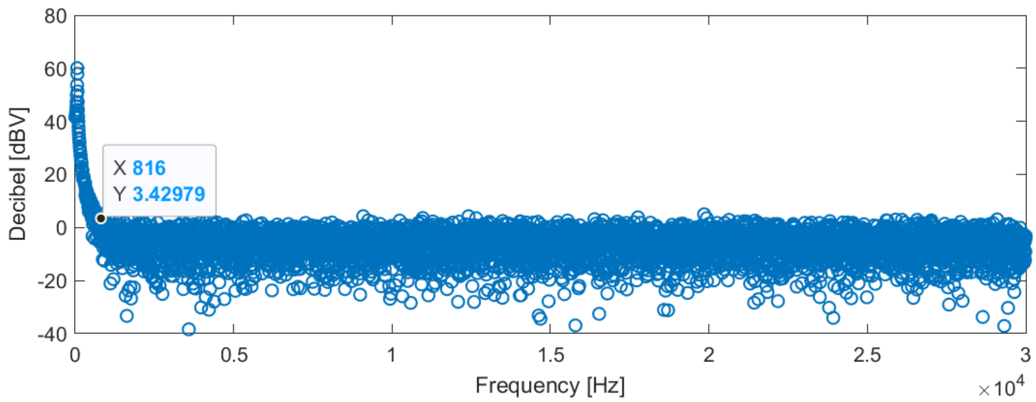


Figure 2.3: Output spectrum - multisine up to 30 KHz

A careful analysis of the output spectrum reveals a rapid decay beyond approximately 100 Hz, followed by a plateau starting around 800 Hz. Beyond 800 Hz, the contribution of the system dynamics and nonlinear effects becomes negligible, and the measured output is dominated by the noise floor, which is assumed to be approximately white.

From these observations, we decided to use a sampling frequency (f_s) of 8 kHz for the BLA Frequency response estimation. The excited frequency band is limited to 800 Hz with the goal to capture the nonlinear effects while maintaining a sufficient signal-to-noise ratio and avoiding aliasing. It is worth mentioning that measurement system contains a **zero-order-hold** block, convolving the discrete samples with a square pulse, which is equivalent to a multiplication, in the frequency domain, of the input spectrum with a sinc function centered in 0 Hz. This multiplication shape the input spectrum with a magnitude reduction due to the sinc-lobe envelope. In our case, this magnitude degradation is limited as the excited band is restricted in the range from 0 Hz (not included) to 800 Hz, sufficiently far to limit this side effect - Figure 2.4.

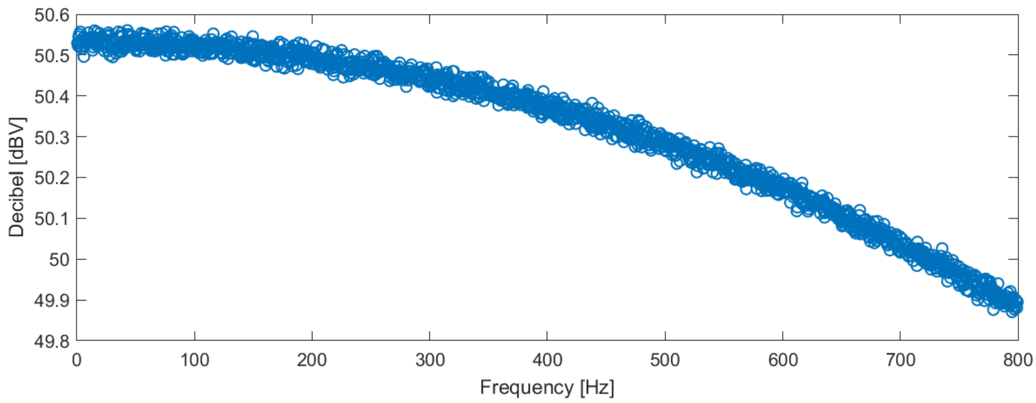


Figure 2.4: Zero-order-hold - sinc-shaped magnitude response

Finally, as we are working with multisine excitations with multiple periods, the first period is systematically removed to eliminate the transient which would affect the BLA estimation - Figure 2.5.

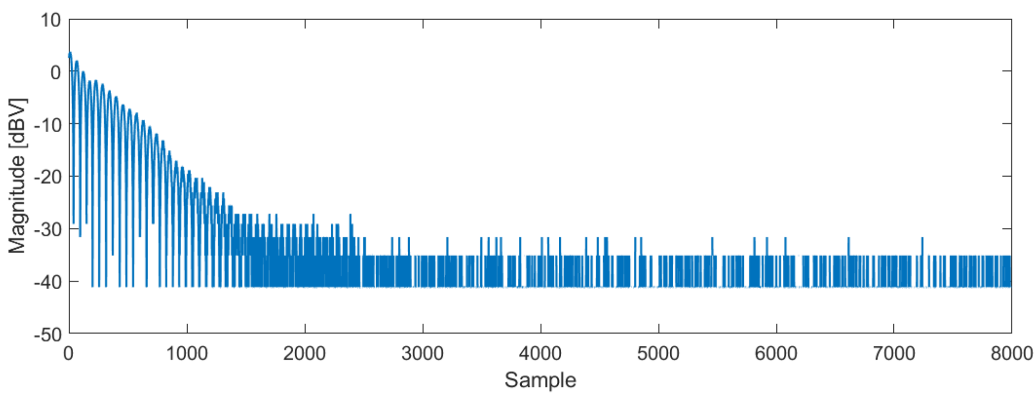


Figure 2.5: Transient response removal

2.4 BLA Frequency Response estimation methods

2.4.1 Fast method

The first method, namely **Fast method**, arises from the observation that a multisine input exciting frequencies at odd multiples of the fundamental frequency results in responses where odd nonlinear contributions appear at odd output frequencies and even nonlinear contributions will appear at even output frequencies. From this property, we can construct a multisine exciting odd multiples of the fundamental frequency with random holes. The generated output contains :

1. **Linear contributions** at excited frequencies.

2. **Even nonlinear contributions** at even multiples of fundamental frequency.

3. **Odd nonlinear contributions** at non excited odd frequencies (holes location).

A simple trick is used to extract the noise contributions by adding multiple periods in order to increase the frequency resolution and "add" bins in the DFT grid (valid for PISPO systems).

Figure 2.6 shows the estimated linear, nonlinear and noise contributions for the system under evaluation. The observed nonlinear distortion is confined to odd harmonics (approximately 22 dB lower than the linear contributions), which is consistent with the assumption made in the *System and measurement devices* section regarding the presence of cubic nonlinearity in the static feedback. The even nonlinear contributions remain at the noise level.

It should be highlighted that the extraction of the noise contributions shown in Figure 2.6 is made possible by setting the fundamental excited frequency at the third frequency bin. This choice allows the noise contribution to be estimated from a single period experiment. However, the odd and even nonlinear contributions are evaluated over a limited number of frequency bins, resulting in a reduced amount of information compared to methods that rely on increasing the frequency resolution by capturing multiple periods. Were the experiment to be redone, the input signal would be designed to follow the latter method leading to a more detailed estimation of the nonlinearity.

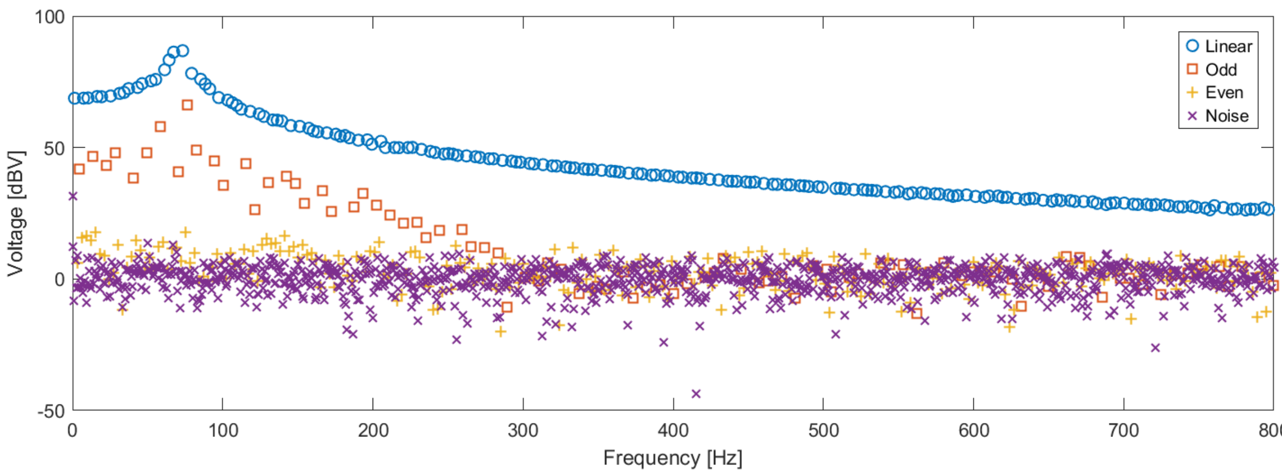


Figure 2.6: Output response spectrum - Fast Method

From the output spectrum, we can compute an estimation of the BLA from the output spectrum but also the input signal-dependent nonlinear gain by dividing the nonlinear contributions with the excited input at its neighborhood - Figure 2.7.

$$Y(k_e) = Y_{\mathrm{BLA}}(k_e) + Y_s(k_e), \quad \forall k_e \in K_{\mathrm{exc}}, \tag{2.5}$$

$$G_{\mathrm{BLA-fast}}(k_e) = \frac{Y(k_e)}{U(k_e)}, \tag{2.6}$$

$$G_{\mathrm{NL-fast}}(k_{ne}) = \frac{Y_s(k_{ne})}{U(k_e)}, \qquad \forall k_{ne} \in K_{\mathrm{non-exc}} \tag{2.7}$$

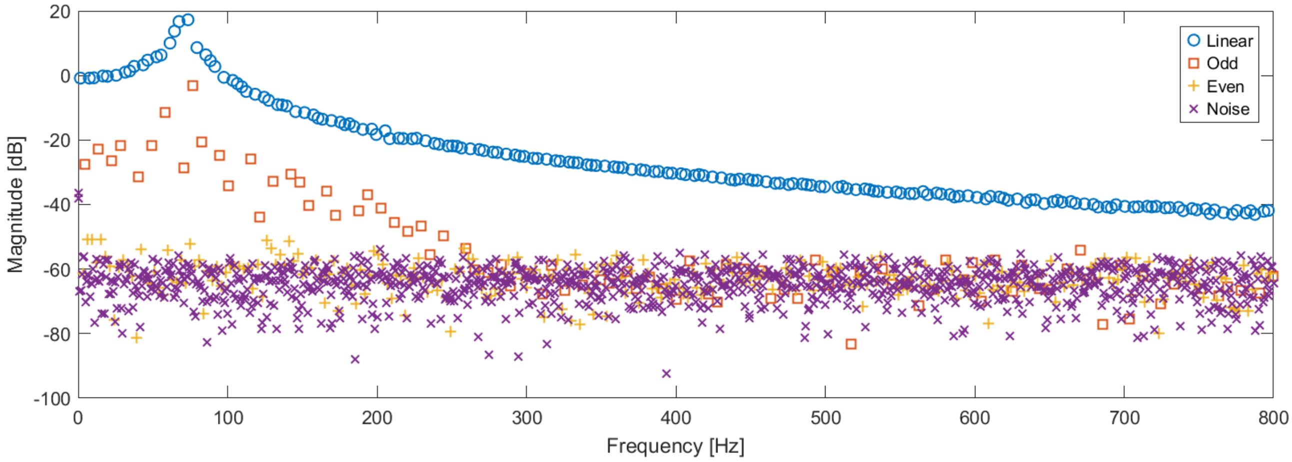


Figure 2.7: G_{BLA} and nonlinear gain estimation - Fast Method

2.4.2 Robust method

By carefully examining the output spectrum in Figure 2.6, we can notice that the nonlinear distortion presents a noisy behaviour. This stochastic evolution results from the generation of sines with random phase at each excited frequency. Consequently, an improved methodology can be proposed by averaging the output response over multiple realisations. This alternative approach, namely the **Robust method**, averages out the non-coherent nonlinear contributions, $Y_s(k)$, yielding a more accurate estimate of the frequency response function of the BLA, while also allowing the computation of the power spectrum of the nonlinear distortions - as described in Equations 2.9 and 2.10

$$\mathbb{E}\{Y_s(k)\} = 0, \tag{2.8}$$

$$\mathbb{E}\{Y_s(k_e)U(k_e)\} = 0, \tag{2.9}$$

$$\mathbb{E}\{|Y_s(k)|^2\} = \sigma_s^2(k). \tag{2.10}$$

The Robust method consists of exciting the system with a random-phase multisine input (with all frequency bins excited). This input signal is repeated over P periods and a new random-phase multisine is generated for each of the M realizations. Thus, the noise variance can be estimated over the periods, while the nonlinear variance is computed across the realizations. As previously mentioned, the smoother estimate of the frequency response function of the BLA is obtained through the equations 2.11, 2.12 and 2.13.

$$\hat{G}^{[m,p]} = \frac{Y^{[m,p]}}{U^{[m]}}, \tag{2.11}$$

$$\hat{G}^{[m]} = \frac{1}{P}\sum_{p=1}^{P}\hat{G}^{[m,p]}, \tag{2.12}$$

$$\hat{G}_{\text{BLA}} = \frac{1}{M}\sum_{m=1}^{M}\hat{G}^{[m]}. \tag{2.13}$$

The sample mean variance of the noise is calculated over the P periods - Equation 2.14. The normalisation of the mean variance by $P(P-1)$ arising from substituing the expected value by the average sample and from considering the variance of the sample mean. The sample mean is, then, refines over the M realisations - Equation 2.15. The sample mean variance of the BLA FRF, computed over the realisation, is affected by the nonlinear distortion and the noise - Equation 2.16. The sample mean variance of the nonlinear distortion can be computed by subtracting the noise variance from the sample mean variance of the BLA.

$$\hat{\sigma}_n^{2[m]} = \frac{1}{P(P-1)}\sum_{p=1}^{P}|\hat{G}^{[m,p]} - \hat{G}^{[m]}|^2, \tag{2.14}$$

$$\hat{\sigma}_n^2 = \frac{1}{M^2}\sum_{m=1}^{M}\hat{\sigma}_n^2[m], \tag{2.15}$$

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{M(M-1)}\sum_{m=1}^{M}\left|\hat{G}^{[m]} - \hat{G}_{\text{BLA}}\right|^2. \tag{2.16}$$

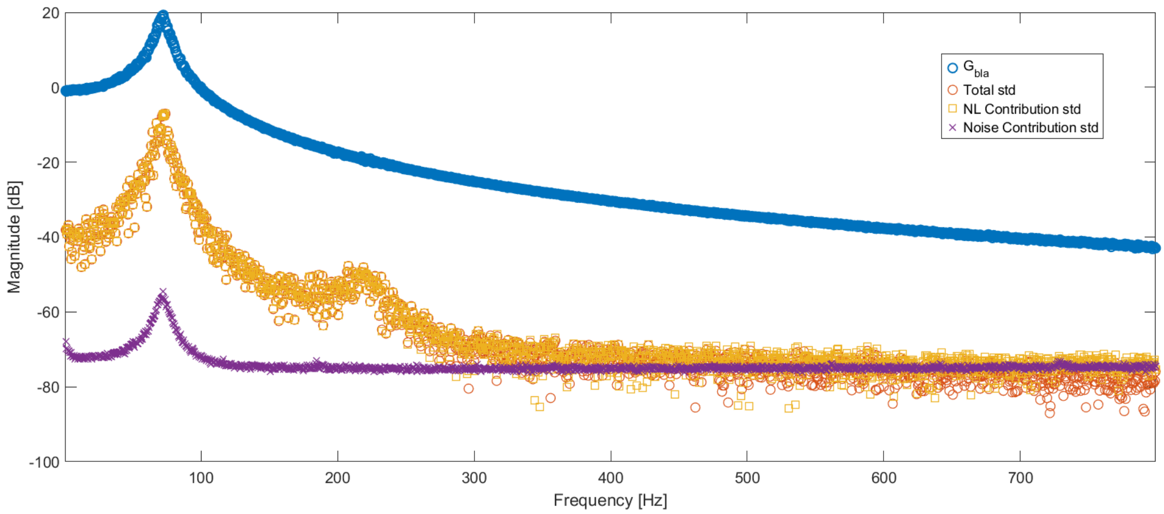


Figure 2.8: G_{BLA} estimation and associated uncertainty level - Robust Method

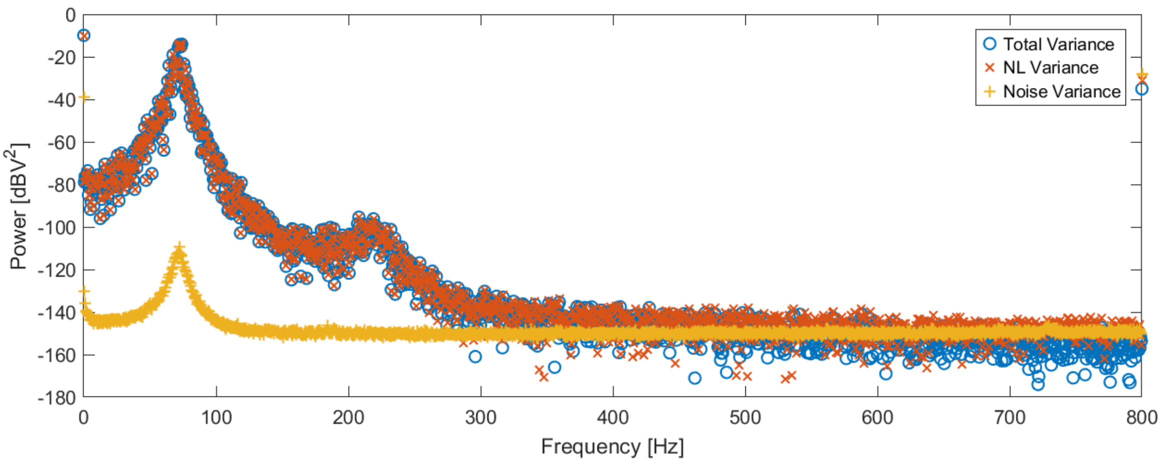


Figure 2.9: BLA, Nonlinear distortion and noise variances - Robust Method

The normalisation factor results to a G_{BLA} uncertainty reduction of a factor $\sqrt{M(M-1)}$ compared to the uncertainty level computed with the fast method. The *robust* experiment contains 10 periods (P) and 18 realisations (M) and lead to an uncertainty reduction of 12.4 dB. It can be verified by comparing the uncertainty computed with the Fast method (Figure 2.7) and with the Robust method (Figure 2.8) leading to a difference of ∼ 12 dB.

2.4.3 Bonus - System in |V|V^2 feedback mode

The reader can find the results of the Fast and Robust method when the feedback actuator is positioned on |V|V^2 assuming even and odd nonlinear distortions. The results are given as a bonus and only the system with cubic feedback will be studied in the next chapter regarding the parametric model estimation.

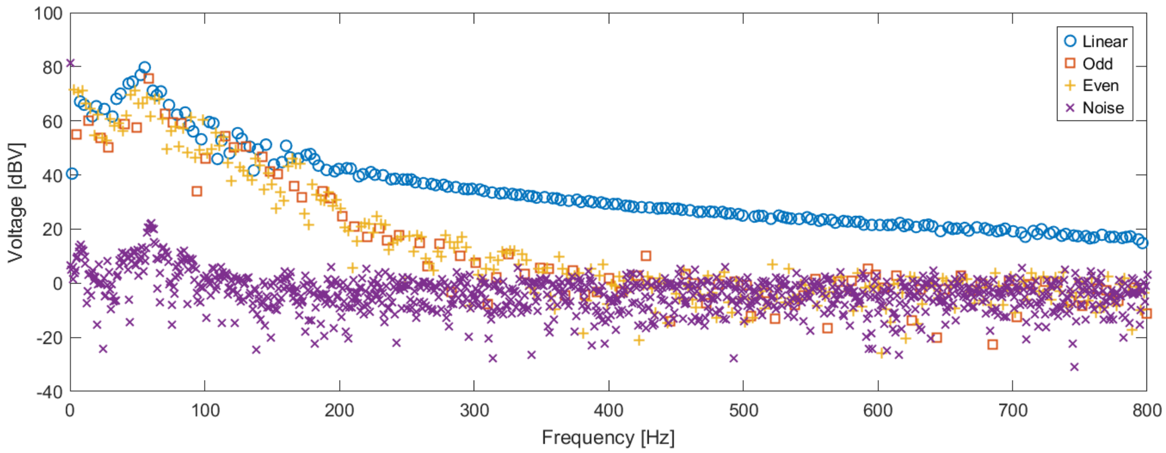


Figure 2.10: Output response spectrum - Fast Method - |V|V^2 mode

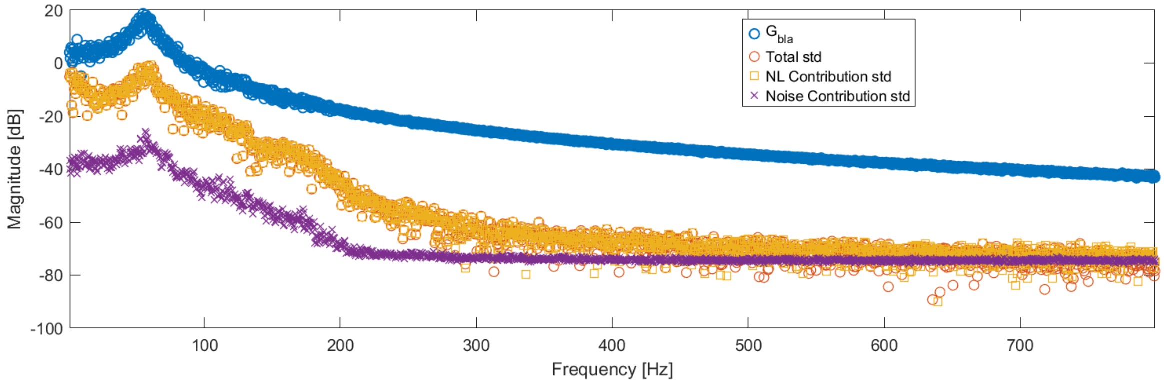


Figure 2.11: G_{BLA} estimation and associated uncertainty level - Robust Method - $|V|V^2$ mode

3
Parametric model estimation

The previous chapter explained the methodology to acquire and compute the **Best Linear Approximation FRF** and its variance over the frequencies of the band of interest. We have now the necessary information to estimate the rational form of the underlying transfer function - Equation 3.1.

$$G(\Omega,\theta) = \frac{\sum_{r=0}^{n_b} b_r \Omega^r}{\sum_{r=0}^{n_a} a_r \Omega^r} \tag{3.1}$$

Multiple estimators can retrieve the parameters with different levels of uncertainty associated to the estimator cost function. The asymptotically optimal estimator is the **Maximum likelihood estimator** (ML) leading to the cost function being minimal in the true parameters as the number of sample tends to infinity 3.3.

$$V_{\mathrm{ML}}(\theta, Z) = \sum_k \frac{|e(\Omega_k, \theta, Z(k))|^2}{\sigma_e^2(\Omega_k, \theta)} \tag{3.2}$$

$$V_{\mathrm{F}}(\theta) = \mathbb{E}\{\frac{1}{F} V_{\mathrm{ML}}(\theta, Z)\} = \mathbb{E}\{\frac{1}{F}\sum_{k=1}^{F} \frac{|e(\Omega_k, \theta, Z_0(k))|^2}{\sigma_e^2(\Omega_k, \theta)}\} + 1 \tag{3.3}$$

The main drawback of such estimator lies in the fact that the cost function to minimize is nonlinear in the parameters resulting into a non-convex problem. The trick, then, is to approximate the ML estimator using linear least squares convex relaxation with well chosen constraint on the parameters. The chapter will present the derived linear estimators used for estimating the transfer function parametric model and attend to explain the limitations and give a critical analysis based on the results. Additionally, we will introduce more advanced algorithms giving estimation of the ML estimator. Finally, the chapter will conclude by selecting the most suitable model order through the **Bayesian Information Criteria** (BIC) algorithm.

3.1 Solving bad numerical conditioning of the regression matrix (J)

In the following sections, we will explain how are designed the estimators and make multiple mentions of the **regression matrix** (J). This matrix is constructed such that each of its columns corresponds to a basis formed from the inputs and past outputs. The bases are mutliplied by the powers of jw_k making J frequency dependent. This multiplication with jw_k powers becomes problematic especially if the number of parameters, n_θ, increases leading to a poor numerical conditioning of J. To mitigate this problem, a simple trick is to normalize the columns of J by their corresponding root mean squared value (RMS) - Equation 3.4.

$$S = \begin{bmatrix} \frac{1}{\mathrm{RMS}(J_{[:,1]})} & 0 & 0 & \cdots & 0 \\ 0 & \frac{1}{\mathrm{RMS}(J_{[:,2]})} & 0 & \cdots & 0 \\ 0 & 0 & \frac{1}{\mathrm{RMS}(J_{[:,3]})} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \frac{1}{\mathrm{RMS}(J_{[:,n]})} \end{bmatrix} \tag{3.4}$$

The change of variable is expressed as :

$$y = J\theta \quad \Longleftrightarrow \quad y = \tilde{J}\tilde{\theta} \tag{3.5}$$

$$\text{with} \quad \tilde{J} = JS \quad \text{and} \quad \tilde{\theta} = S^{-1}\theta. \tag{3.6}$$

3.2 Linear Least Squares estimator (LLS)

The first linear relaxation of the ML estimator we will discuss is the **Linear Least Squares** estimator. The cost function is reduced to the minimization of the numerator of the ML estimator - Equation 3.7 and finding the parameters a_n and b_n that minimize the error - Equation 3.8.

$$V_{\mathrm{LLS}}(\theta, Z) = \sum_{k=1}^{F} |e(\Omega_k, \theta, Z(k))|^2 \tag{3.7}$$

$$e(\theta, \Omega_k) = Y(\Omega_k) - G(\theta, \Omega_k)\, U(\Omega_k) \tag{3.8}$$

By multiplying the expression with $A(\theta, \Omega_k)$, the equation becomes linear in the parameters and is rewritten as :

$$e(\theta, \Omega_k) = A(\theta, \Omega_k)\, Y(\Omega_k) - B(\theta, \Omega_k)\, U(\Omega_k) \tag{3.9}$$

$$J = \begin{bmatrix} -Y(\omega_1) & -Y(\omega_1)\,\Omega_1 & -Y(\omega_1)\,\Omega_1^2 & \cdots & U(\omega_1) & U(\omega_1)\,\Omega_1 & U(\omega_1)\,\Omega_1^2 & \cdots \\ -Y(\omega_2) & -Y(\omega_2)\,\Omega_2 & -Y(\omega_2)\,\Omega_2^2 & \cdots & U(\omega_2) & U(\Omega_2)\,\Omega_2 & U(\Omega_2)\,\Omega_2^2 & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots \\ -Y(\omega_N) & -Y(\omega_N)\,\Omega_k & -Y(\omega_N)\,\Omega_N^2 & \cdots & U(\omega_N) & U(\omega_N)\,\Omega_N & U(\omega_N)\,\Omega_N^2 & \cdots \end{bmatrix} \tag{3.10}$$

$$\theta = \begin{bmatrix} a_0 & a_1 & a_2 & \cdots & a_{na} & b_0 & b_1 & b_2 & \cdots & b_{nb} \end{bmatrix}^{\mathrm{T}} \tag{3.11}$$

Leading to the final expression :

$$e(\theta, \Omega_k) = J(\Omega_k)\,\theta \tag{3.12}$$

We can notice that the trivial solution $\mathrm{a_n} = 0$ and $\mathrm{b_n} = 0$ is part of the problem solution making the regression matrix rank deficient. By enforcing $\mathrm{a_0} = 1$, the problem becomes well-posed and ensures that the residual is minimized by finding meaningful values which still results in inconsistent estimate (noise weight is considered uniform over the frequencies for LLS) - Equation 3.13.

$$Y(\Omega_k) = J(\Omega_k)\,\theta_{\mathrm{LLS}} \tag{3.13}$$

$$\hat{\theta}_{\mathrm{LLS}} = J^{\dagger}\,Y \tag{3.14}$$

Incorporating the nonparametric FRF estimate into the regression matrix

By cleverly setting the input $U(\Omega_k) = 1$,we can make appear the G_{BLA} estimate computed with the **Robust Method** in J as follow :

$$Y(\Omega_k) = G_{\mathrm{BLA}}(\Omega_k) \tag{3.15}$$

Imposing real parameters

Before exploring more advanced linear estimators, we should pause to discuss the nature of the parameters. The parameters are computed from the minimization of errors which are complex and thus, may lead to generation of complex paramater. However, we estimate during this project a system which is real and, therefore, characterized by a transfer function with **real** components. We can impose the real nature of the parameters by concatenating the real and imaginary part of J such that :

$$J_{\mathrm{re}} = \begin{bmatrix} \mathrm{Re}\{J\} \\ \mathrm{Im}\{J\} \end{bmatrix} \tag{3.16}$$

$$\mathrm{Re}\,\hat{\theta}_{\mathrm{LLS}} = J_{\mathrm{re}}^{\dagger}\,Y \tag{3.17}$$

3.3 Total Least Squares estimator (TLS)

An alternative method to remove the trivial solution of being part of the solution space is to constraint the parameters such that $\|\theta\|_2^2 = 1$. This estimator will still result in inconsistent estimates but is more robust to ill-problem.

$\hat{\theta}_{\text{TLS}}$ is obtained from the Singular Value Decomposition of J :

$$J = U_J \Sigma_J V_J^{\text{H}} \tag{3.18}$$

$$\hat{\theta}_{\text{TLS}} = V_{J_{[:,\text{end}]}} \tag{3.19}$$

Selecting the last column ensures $\hat{\theta}_{\text{TLS}}$ to be the singular vector corresponding to the smallest singular value.

3.4 Generalized Total Least Squares estimator (GTLS)

GTS is an extension of the TLS estimator with a more appropriate choice of the parameters constraint leading to consistent estimate.

$$V_{\text{GTLS}}(\theta) = \|J\theta\|_2^2 \quad \text{subject to} \quad \|C_{\text{J}}^{\frac{1}{2}}\theta\|_2^2 = 1 \tag{3.20}$$

Where C_{J} is the column covariance of J. It is computed as :

$$C_{\text{J}} = \mathbb{E}\{\Delta J^{\text{H}} \Delta J\} \tag{3.21}$$

This constraint formulation imposes a weighting on J by C_{J}. By imposing $\|C_{\text{J}}^{\frac{1}{2}}\theta\|_2^2 = 1$, the expected value of the noisy part of the cost function V_{GTLS} remain θ-independent, resulting to a minimization of the expectated value of the cost function in the true parameters.

C_{J} computation

The following subsection will explicitly explain the computation of the column covariance of J.

By imposing that $U(\Omega_k) = 1$, the noisy part of the regression matrix, ΔJ, can be computed as:

$$\Delta J = \begin{bmatrix} \vdots & & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & & \vdots \\ N_{\text{GBLA}}(\omega_k) & \cdots & \Omega_k^{n_a} N_{\text{GBLA}}(\omega_k) & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & & \vdots \end{bmatrix} \tag{3.22}$$

Where the input noise is null as the input signal is set to 1 thus deterministic.

$$C_{\mathrm{J}} = \begin{bmatrix} \sum_k \sigma_y^2(\omega_k) & \sum_k \sigma_y^2(\omega_k)\,\Omega_k & \sum_k \sigma_y^2(\omega_k)\,\Omega_k^2 & \cdots & 0 & \cdots & 0 \\ \sum_k \sigma_y^2(\omega_k)\,\Omega_k^* & \sum_k \sigma_y^2(\omega_k)\,|\Omega_k|^2 & \vdots & & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ 0 & 0 & 0 & & 0 & & 0 \end{bmatrix} \tag{3.23}$$

Once again, to ensure that the parameters are real, we have to take the real part of the column variance matrix :

$$C_{J_{re}} = \mathrm{Re}\{C_J\} \tag{3.24}$$

Parameters computation

The Equation 3.20 can be rewritten as :

$$V_{\mathrm{GTLS}}(\theta) = \|C_J^{-1/2} J\tilde{\theta}\|_2^2 \quad \text{subject to} \quad \|\tilde{\theta}\|_2^2 = 1 \tag{3.25}$$

Considering that $C_{\mathrm{J}}^{\frac{1}{2}}$ is singular, finding the estimated parameters is done through the **Generalised Singular Value Decomposition** of the matrix pair $(\tilde{\mathrm{J}}, C_{\tilde{\mathrm{J}}}^{\frac{1}{2}})$:

$$\tilde{\mathrm{J}} = UCX^T \tag{3.26}$$

$$C_{\tilde{\mathrm{J}}}^{\frac{1}{2}} = VSX^T \tag{3.27}$$

It remains to find the index i which minimize the ratio of singular values $\frac{c_i}{s_i}$ to compute the estimated parameters :

$$\hat{\theta}_{\mathrm{GTLS}} = \left(X^T\right)^{-1}_{[:,i]} \frac{1}{s_i} \tag{3.28}$$

It is worth mentioning that, from the equation 3.6, the noisy part of J, ΔJ, is also normalized by the RMS of the regression matrix columns.

3.5 Bootstrapped Total Least Squares estimator (BTLS)

The BTLS estimator is an extension of the GTLS and addresses the issue of overemphasizing uncertain frequencies observed in previous linear estimators by weighting the frequencies appropriately based on the variance error associated on that specificaly frequency. The GTLS cost function is updated with the weighted regression matrix :

$$\mathrm{J}_{\mathrm{re\_BTLS}} = W\,\mathrm{J}_{\mathrm{re}}\,S \tag{3.29}$$

With W denoted as the diagonal weighted matrix :

$$W = \mathrm{diag}\left(\sigma_e^{-r}\left(\Omega_k, \hat{\theta}^{(i-1)}\right)\right) \tag{3.30}$$

$$\sigma_e^2(\Omega_k, \hat{\theta}^{(i)}) = |A(\Omega_k, \hat{\theta}^{(i)})|^2 \, \sigma^2_{\mathrm{G_{BLA}}} \tag{3.31}$$

With the error variance, σ_e^2, computed with the previous parameters estimate. The algorithm being iterative, we would need a first estimate of the parameters to initialize the computation - in our case the GTLS estimate is our starting point. One can also observe that, by fixing r = 1, the cost function tends to the formulation of the Maximum Likelihood - this r value (r = 1) is used for the calculation of the parameters estimate with BTLS.

$$V_{\mathrm{BTLS}}(\theta^{(i)}, Z) = \sum_k \frac{|e(\Omega_k, \theta^i, Z(k))|^2}{\sigma_e^{2r}(\Omega_k, \theta^{(i-1)})} \tag{3.32}$$

To ensure the algorithm is not trapped into an infinity loop, we defined two stopping criteria. The first one is a maximum number of iterations we allowed. The second one is a comparison between the cost newly computed and the previous one. The algorithm stops once the computed cost is bigger than the previous one or if the maximum number of iterations has been reached.

3.6 Nonlinear optimizers

We will now look at optimizers that allow nonlinear, non quadratic cost function. These optimizers will be used to provide numerical solution for the Maximimum Likelihood cost function which, as remainder, is the asymptotically optimal estimator. Those optimizers are iterative algorithms which required an initial value for the parameter. The difference with BTLS lies in the fact that BTLS computes the weights based on the previous parameters estimate, and minimize a quadratic cost function.

3.6.1 Newton-Gauss

First, we need to express the terms that will be used in the optimizer :

$$V_{\mathrm{ML}}(\theta, Z) = \frac{1}{2} \sum_k \frac{|e(\Omega_k, \theta, Z(k))|^2}{\sigma_e^2(\Omega_k, \theta)} \tag{3.33}$$

$$\text{We impose} \quad \epsilon_{[k]}^{(i)} = \frac{e(\Omega_k, \theta^{(i)}, Z(k))}{\sigma_e^2(\Omega_k, \theta^{(i)})} \tag{3.34}$$

$$\mathscr{J}_{[k,r]}^{(i)} = \frac{\partial \epsilon_{[k]}}{\partial \theta_{[r]}^{(i)}} \tag{3.35}$$

The Newton-Gauss optimizer relies on the Newton-Raphson approach :

$$\mathrm{F}(\theta^{(\mathrm{i})})\,\Delta\theta^{(i+1)} = -\mathrm{F}(\theta^{(i)}) \tag{3.36}$$

$$\text{Considering that} \quad F(\theta^{(i)}) = \frac{\mathrm{d}V}{\mathrm{d}\theta} = \frac{\mathrm{d}\epsilon^{(i)H}}{\mathrm{d}\theta}\,\epsilon^{(i)} = \mathscr{J}^{(i)H}\,\epsilon^{(i)} \tag{3.37}$$

$$\text{Leading to} \quad \frac{\mathrm{d}(\mathscr{J}^{(i)H}\,\epsilon^{(i)})}{\mathrm{d}\theta}\,\Delta\theta^{(i+1)} = -\mathscr{J}^{(i)H}\,\epsilon^{(i)} \tag{3.38}$$

$$\text{Finally} \quad (\mathscr{J}^{(i)H}\,\mathscr{J}^{(i)} + \frac{\mathrm{d}\mathscr{J}^{(i)H}}{\mathrm{d}\theta}\,\epsilon^{(i)})\,\Delta\theta^{(i+1)} = -\mathscr{J}^{(i)H}\,\epsilon^{(i)} \tag{3.39}$$

Newton-Gauss assumes that the error, $\epsilon^{(i)}$, is approximatevely equals to 0 for good initial estimate computed from the linear estimators (TLS, GTLS or BTLS). The Equation 3.40 is therefore reduced to :

$$(\mathscr{J}_{re}^{(i)H}\,\mathscr{J}_{re}^{(i)})\,\Delta\theta^{(i+1)} = -\mathscr{J}_{re}^{(i)H}\,\epsilon_{re}^{(i)} \tag{3.40}$$

$$\Delta\theta^{(i+1)} = -\mathscr{J}_{re}^{(i)\dagger}\,\epsilon_{re}^{(i)} \tag{3.41}$$

$$\text{with} \quad \epsilon_{re} = \begin{bmatrix} \mathrm{Re}\{\epsilon\} \\ \mathrm{Im}\{\epsilon\} \end{bmatrix} \tag{3.42}$$

Where the newly estimated parameters are the sum of the previous estimate and the $\Delta\theta$ computed in 3.41.

In order to make the $\Delta\theta$ computation numerically stable, the pseudo-inverse can be computed through singular value decomposition :

$$\Delta\theta^{(i+1)} = -\mathscr{J}_{re}^{(i)\dagger}\,\epsilon_{re}^{(i)} \tag{3.43}$$

$$= V^{(i)}\,\Sigma^{(i)\dagger}\,U^{(i)\,T}\,\epsilon_{re}^{(i)} \tag{3.44}$$

The regressor *close*-singularity (rank deficient) is handled by truncating the singular value associated with the unidentifiable direction (the smallest singular value set to 0).

In summary, the algorithm is iteratively updating the parameters. The number of iteration has been set manually to 20 in our case.

Computation of the polynomials (A_k and B_k) derivatives

The section will not present in details the computation of the jacobian, ($\mathscr{J}$), but will express the polynomials (A_k and B_k) derivatives that appears in the mathematical expression of the jacobian explained in the course of *Identification of dynamic systems.* Let's first recall the error expression from 3.9 :

$$e(\theta,\Omega_k) = A(\theta_a,\Omega_k)\,G_{BLA}(\Omega_k) - B(\theta_b,\Omega_k) \tag{3.45}$$

The derivatives of the error with respect with the θ can be expressed as :

$$\frac{\partial A_k}{\partial \boldsymbol{\theta}} = \begin{bmatrix} 1 & \Omega_k & \cdots & \Omega_k^{n_a} & \mathbf{0}_{[1\times(n_b+1)]} \end{bmatrix} \tag{3.46}$$

$$\frac{\partial B_k}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \mathbf{0}_{[1\times(n_a+1)]} & 1 & \Omega_k & \cdots & \Omega_k^{n_b} \end{bmatrix}. \tag{3.47}$$

Resulting to :

$$\frac{\partial e_k}{\partial \boldsymbol{\theta}} = \begin{bmatrix} G_{BLA}(\Omega_k) & \Omega_k G_{BLA}(\Omega_k) & \cdots & \Omega_k^{n_a} G_{BLA}(\Omega_k) & -1_{[N\times 1]} & -\Omega_k & \cdots & -\Omega_k^{n_b} \end{bmatrix} \tag{3.48}$$

3.6.2 Levenberg-Marquardt (LM)

In order to avoid Newton-Gauss to be stuck into a local optimum, Levenberg-Marquardt algorithm introduces an additional term, $\lambda^2 I$ enforcing the convergence by taking a step in the direction of the gradient descent :

$$(\mathscr{J}_{re}^{(i)H} \mathscr{J}_{re}^{(i)} + \lambda^2 I)\,\Delta\theta^{(i+1)} = -\mathscr{J}_{re}^{(i)H} \epsilon_{re}^{(i)} \tag{3.49}$$

Resulting into the computation of $\Delta\theta^{(i+1)}$ through the singular value decomposition :

$$\Delta\theta^{(i+1)} = -V^{(i)}\,\Sigma^{(i)}\,(\Sigma^{(i)2} + \lambda^2)\,U^{(i)\,T}\,\epsilon_{re}^{(i)} \tag{3.50}$$

λ is initialized as $\sigma_{max}(\mathscr{J}_{re})/100$ and updated as follow :

$$\lambda \to 0.4\,\lambda \quad \text{if cost decreases} \tag{3.51}$$

$$\lambda \to 10\,\lambda \quad \text{if cost increase} \tag{3.52}$$

The covariance on the parameters are coming for *free* with the following formula:

$$\mathrm{Cov}(\hat{\theta}) \approx \left(2\,\mathrm{Re}\{J^H J\}\right)^{-1} \tag{3.53}$$

$$= \left(J_{\mathrm{re}}^{\mathrm{T}} J_{\mathrm{re}}\right)^{-1} \tag{3.54}$$

$$= \frac{1}{2}(V\Sigma^{-1})\,(V\Sigma^{-1})^{\mathrm{T}} \tag{3.55}$$

The Equation 3.55 implies that weakly identifiable directions, or equivalently basis vector associated with small singular value, lead to large parameter variances. By linking the uncertainty on the parameters with the uncertainty on the roots - Equation 3.56, we can compute the roots covariance matrix :

$$\delta p = \frac{\mathrm{d}p}{\mathrm{d}\theta}\delta\theta. \tag{3.56}$$

$$\mathrm{Cov}(p) = \frac{\mathrm{d}p}{\mathrm{d}\theta}\mathrm{Cov}(\theta)\left(\frac{\mathrm{d}p}{\mathrm{d}\theta}\right)^H. \tag{3.57}$$

Where the derivative of the roots, $\frac{\mathrm{d}p}{\mathrm{d}\theta}$, is computed as the ratio of the partial derivative of the numerator polynomial (zeros) or denominator polynomial (poles) with respect with the parameters and the partial derivative of the same polynomial with respect with the roots.

3.7 Model Order Selection

In the previous sections of this chapter, we presented the linear estimators we used to get a first approximate of the system transfer function by first fixing the number of poles and zeros randomly. The results of such approach are presented and analyzed in the section *Results - Linear estimators comparison.* Altough this method can help us to observe the robustness and limits of the different estimators, it cannot help us to identify the best parameter space dimension leading to the best estimate of the true underlying transfer function. To reach this aim, we will use the linear estimator as first step for the nonlinear optimizers and rely on a model order selection algorithm to determine the optimal complexity of the model. The results with model order selection are analyzed in the section *Nonlinear optimizers with model order selection* but we first need to introduce the moder order processes used.

Weighted frequency response function error

The first model order procedure relies on finding the parameters dimension which maximize the **log likelihood** between the transfer function estimated with the estimators and the computed G_{BLA} from the system measurements. Mathematicaly, it boils down to :

$$L = \sum_k \frac{|G_{\mathrm{est}}(\Omega_k) - G_{\mathrm{meas}}(\Omega_k)|^2}{\sigma^2_{\mathrm{G_{meas}}}(\Omega_k)} \tag{3.58}$$

Where L denotes the log-likelihood, G_{est} the estimated FRF and G_{meas} the measured FRF.

However,The order selection based on log likelihood tends to select high model order to overfit the data even with the whitening of the errors by the BLA FRF variances. To compensate or penalize this behaviour, we decided to apply the **Bayesian Information Criteria** (BIC) to our estimated data.

Bayesian Information Criteria (BIC)

The particularity of the Bayesian Information Criteria lies in considering not only the maximum value of the likelihood but also the uncertainty of the parameter estimates by looking at the curvature of the likelihood distribution around its optimum. BIC is expressed as the marginal likelihood that the model, or more specifically the model order, has generated the data. Maximizing the Equation 3.59 comes down to finding the best model complexity that generates the data.

$$p(G_{\text{est}}|\mathcal{M}) = \int p(G_{\text{est}}|\theta,\mathcal{M})\,d\theta \tag{3.59}$$

The equation can be rewritten with the Laplace approximation as :

$$\log p(G_{\text{est}}|\mathcal{M}) = \log p(G_{\text{est}}|\hat{\theta},\mathcal{M}) - \frac{1}{2}\log|\mathcal{H}| \tag{3.60}$$

Where $\mathcal{H}$ is the Hessian matrix translating the curvature of the likelihood at the estimated parameter $\hat{\theta}$. We can also express the inverse of the Hessian determinant as a measure of the covariance of the parameter estimates, or equivalently, as the uncertainty on the model parameters. Finally, by simplifying the expression we get :

$$\log p(G_{\text{est}}|\mathcal{M}) = -L(\hat{\theta}) - \frac{d}{2}\log N_{\text{data}} \tag{3.61}$$

Where d denotes as the parameter space dimension and $L(\hat{\theta})$, the likelihood considering the estimated parameter $\hat{\theta}$.

Intuitively, we can understand that increasing the parameter space dimension of an overmodeled system comes down to adding additional parameter direction in the likelihood distribution weakly correlated with the data which results to an increase of the uncertainty and, therefore, a flatter distribution. Such additional dimensions often correspond to redundant dynamics (poles redundancy - pole/zero cancellation), overfitting high frequency or noise-dominant dynamics. BIC penalizes these weakly correlated data directions by constraining the complexity with $d\log N_{\text{data}}$.

3.8 Results

3.8.1 Linear estimators comparison

As previously mentioned, we will review first the model approximations estimated through linear estimators, try to give a critical analyzis of the results to finally move to nonlinear optimizers and look at the poles uncertainty.

Fixed order complexity

Let's first fix the parameter space dimension by setting the numerator and denominator polynomial order to 10 and discuss about the magnitude and phase results for each linear estimator.

By observing the Figures 3.1 and 3.2, one can observe very sharp peaks for LLS and TLS at higher frequencies that the resonance frequency. One possible explanation is that the cost functions of these estimators find their minimum by choosing the parameter vector associated to the minimum singular value of the regressor matrix J (when looking at its singular value decomposition). Possible interpretation is that the curvature of J in the smallest singular value direction is relatively flat and therefore associated to weakly excited dynamics. LLS and TLS tend to exploit these directions to reduce the cost, resulting to an overfit of the high frequencies residuals and, thus, to the sharp peaks at high frequencies.

In the case of the GTLS, the estimator introduces a frequency-dependent weighting based on the uncertainty of the measured FRF. The rows of the regressor is scaled based on the variance at that particular frequency. GTLS will still select the parameter vector associated to the smallest singular value but rescaled based on the FRF uncertainty. Visually, we can indeed notice that the peaks between 50 Hz and 250 Hz, related to dynamics in this frequency range, are downweighted resulting to smoother TF.

The BTLS estimator is iteratively updating the weigthing based on the previous model estimate. The peaks at some frequencies, observed with GTLS, are seen as frequencies with significant variance and is downweigthed at the next iteration leading, in fine, to a smooth transfer function estimate (in phase and in magnitude).

The Figure 3.3 plots the relative magnitude error expressed in decibels :

$$e_{\mathrm{dB}}(\Omega_k) = 20\log_{10}\left(\frac{|G_{\mathrm{est}}(\Omega_k) - G_{\mathrm{BLA}}(\Omega_k)|}{|G_{\mathrm{BLA}}(\Omega_k)|}\right) \tag{3.62}$$

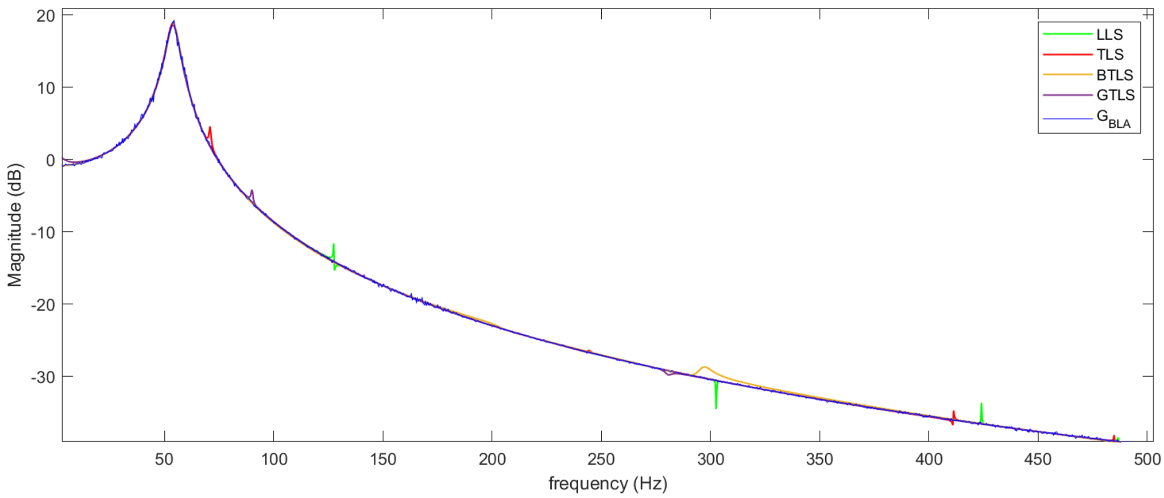


Figure 3.1: Linear Estimators - TF Magnitude estimation - $n_a = 10$ and $n_b = 10$

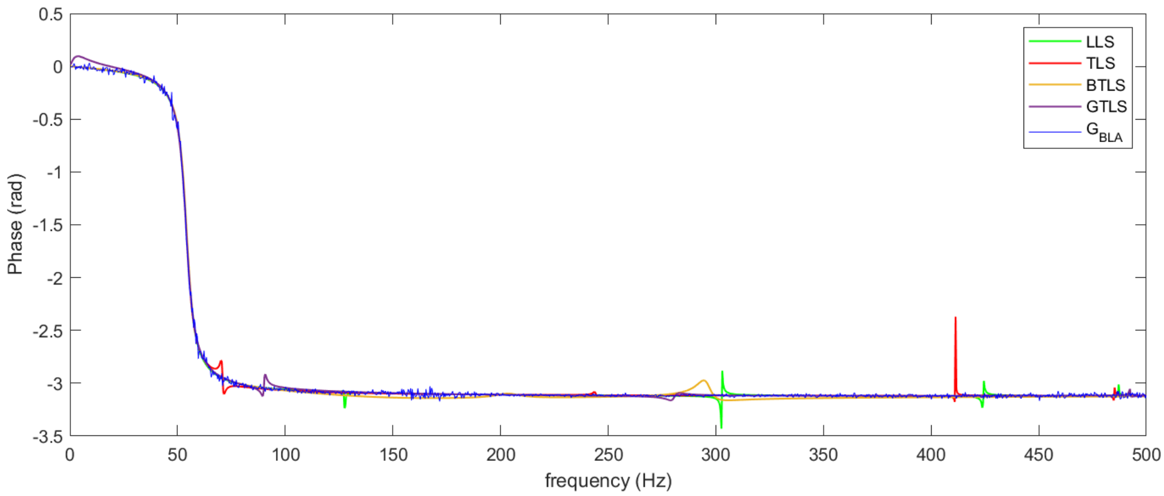


Figure 3.2: Linear Estimators - TF phase estimation - $n_a = 10$ and $n_b = 10$

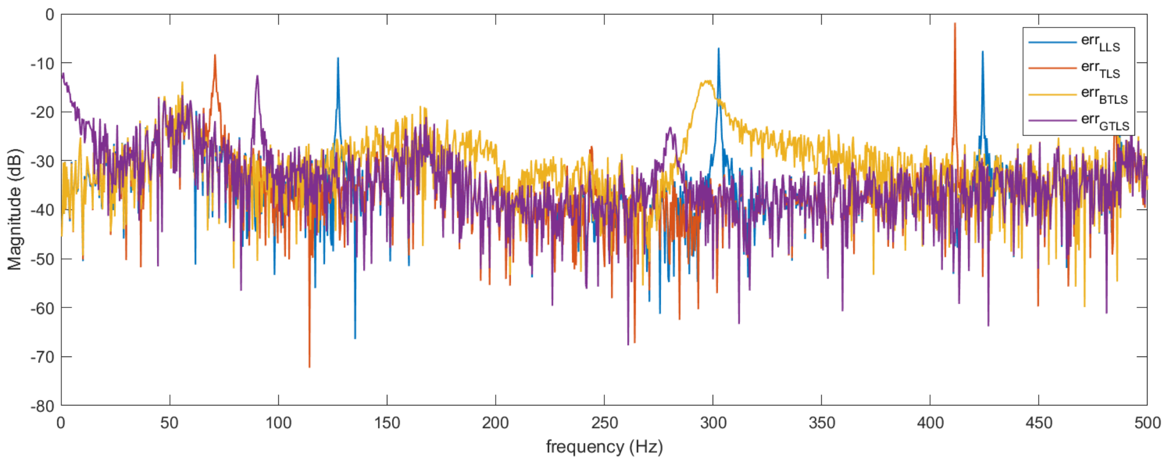


Figure 3.3: Linear Estimators - Error - $n_a = 10$ and $n_b = 10$

BIC model order selection

In the current section, we will now use **BIC** to converge to the most suitable model complexity by penalizing the high order models overfitting noise, distortions or weakly excited dynamics. The results are shown in Figures 3.4, 3.5 and 3.6. It is necessary to mention that the TLS and GLTS curves are coincident which explains the absence of the TLS curves in the figures.

By looking first at the LLS estimates, we can notice a flat estimation of the TF magnitude explained by the single real pole pushed far axay in the negative values. This result can be explained by the penalizing term inside the BIC formulation which privileges such pole. The reason lies in the fact that a large negative real pole affects all the frequency range, fitting globally the TF and therefore a large range of system dynamics. This is made possible as the LLS estimator does not constraint the parameter norm, oppositely to TLS and GTLS which try to fit the oscillatory behaviour of the system (the band where is located the resonance frequency). The BTLS is giving even better results with a very good match of the FRF in magnitude and in phase.

Also, BTLS estimator provides a stable model with limited number of poles (pair of complex poles) located in the left part of the laplacian space - poles : $-22.7 \pm j452.4$. We will retrieve this pair of poles in the models estimated by the nonlinear optimizers which further confirms the robustness of the BTLS estimator with an appropriate estimate initialization. As a reminder, the **r parameter** is set to 1. In contrary, TLS and GTLS proposes a high order model with non-stable poles which results from an overfitting of weakly excited - noisy dynamics.

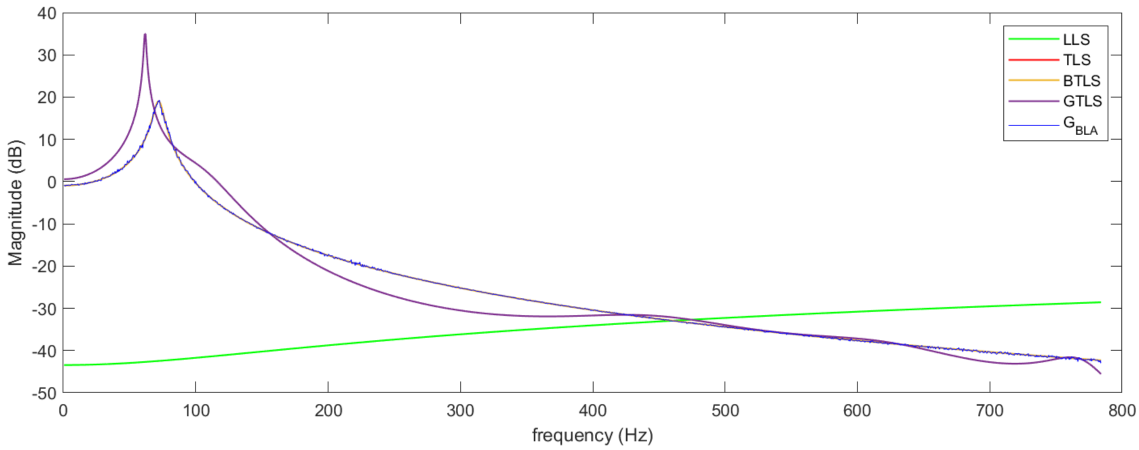


Figure 3.4: Linear Estimators - TF Magnitude estimation - BIC

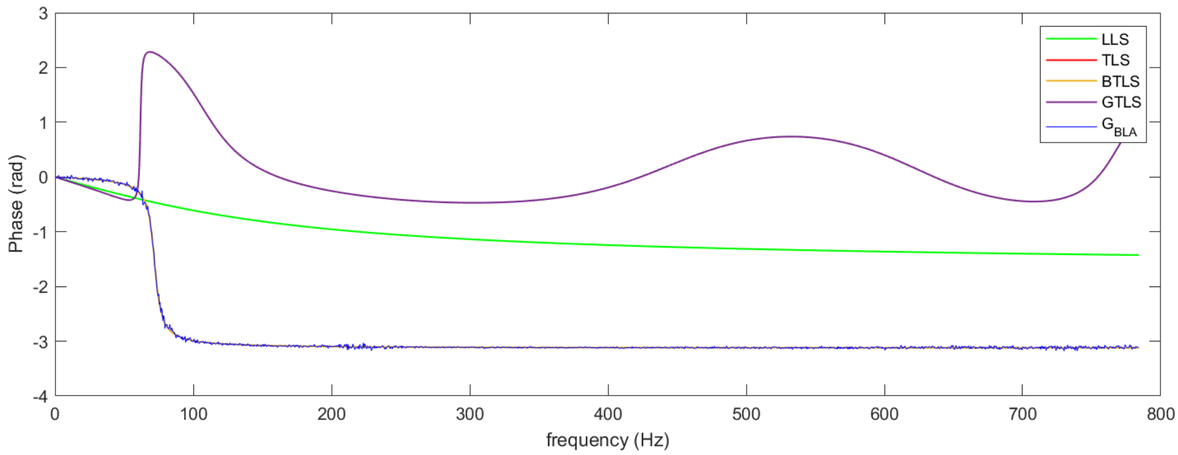


Figure 3.5: Linear Estimators - TF phase estimation - BIC

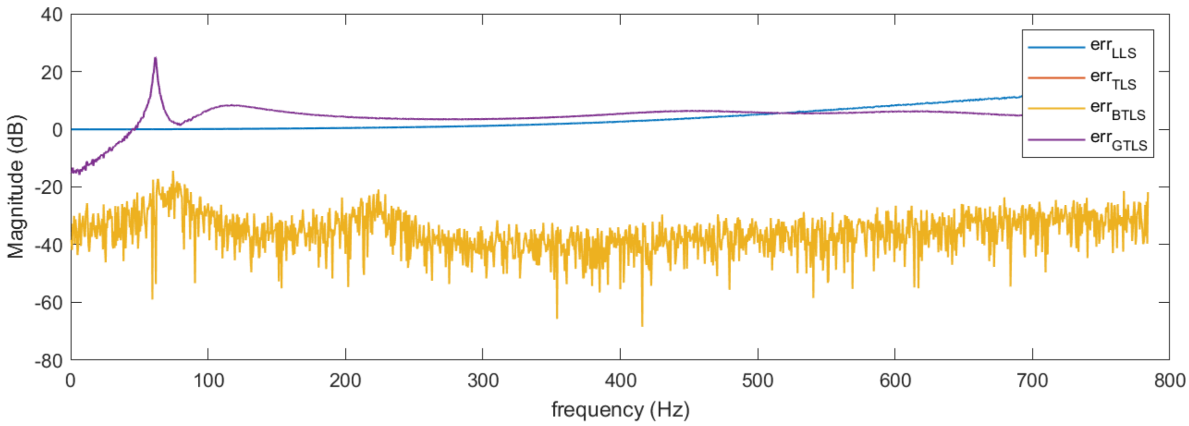


Figure 3.6: Linear Estimators - Error - BIC

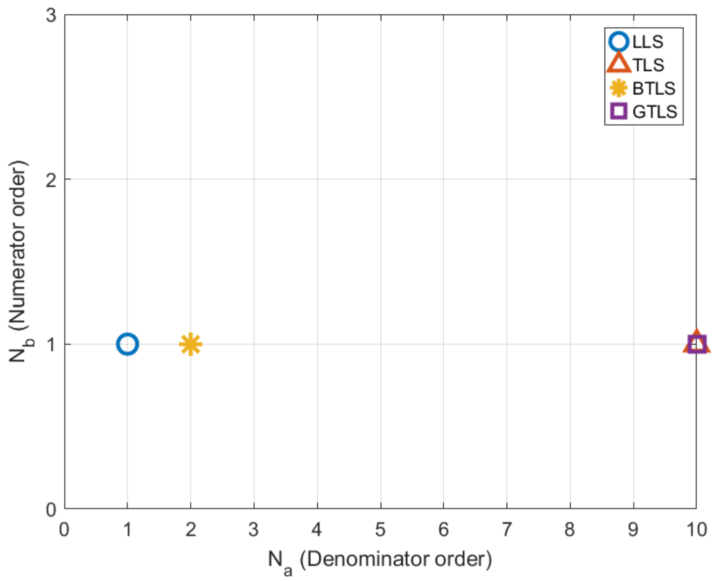
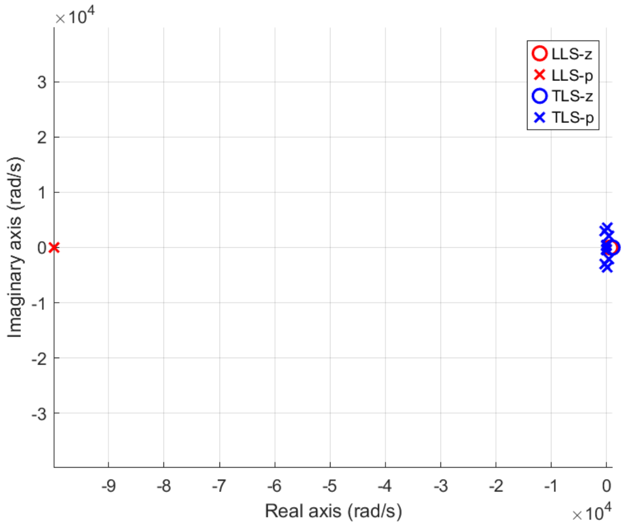
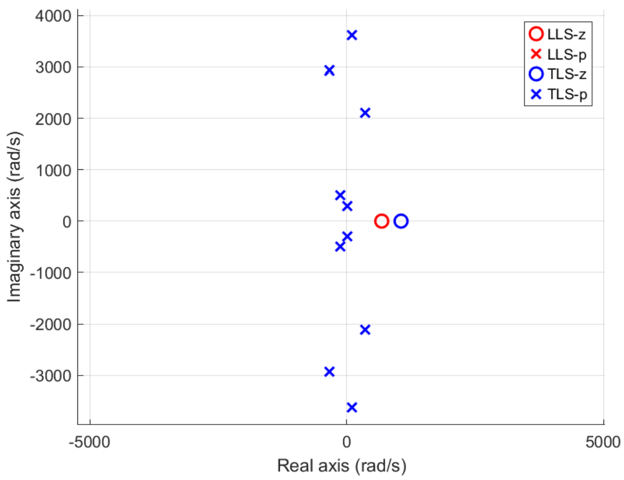


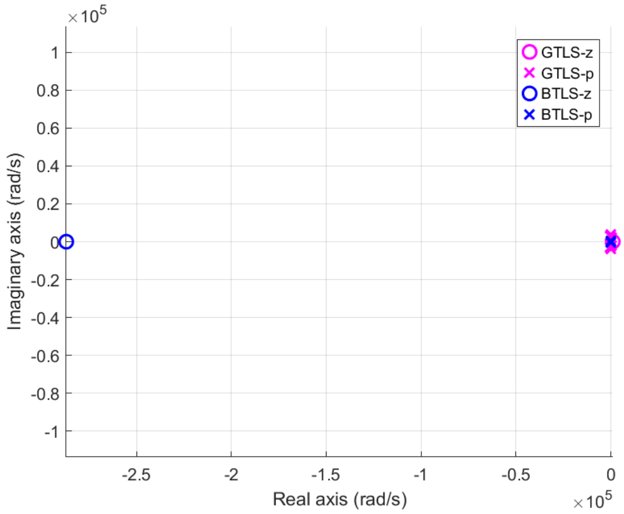
Figure 3.7: Linear Estimators - Parameters space dimension - BIC



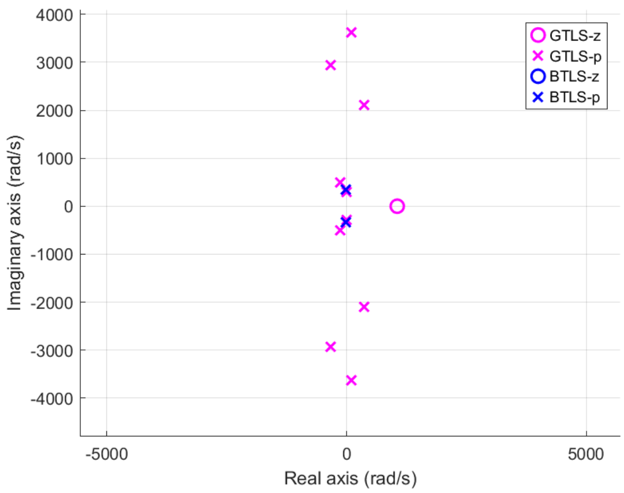
(a) Zeros-Poles Map - LLS and TLS



(b) Zeros-Poles Map Zoom



(c) Zeros-Poles Map - GTLS and BTLS



(d) Zeros-Poles Map Zoom

Figure 3.8: Linear estimators – zeros–poles maps

3.8.2 Nonlinear optimizers with model order selection

We conclude the results section by considering models estimated using nonlinear optimization techniques. The resulting models are compared using different initial estimates obtained from LLS, TLS, GTLS, and BTLS, allowing us to assess the performance improvements associated with initial estimates of varying uncertainty levels.

Newton-Gauss

The first notable result is that, independent of the choice of initial estimate, the nonlinear optimization consistently produces models that accurately capture the magnitude and phase of the transfer function. The second observation concerns the number of poles: compared to the linear estimator models, the number of poles is significantly reduced, indicating that weakly excited dynamics are effectively filtered out. Additionally, Figure 3.11 shows that initial estimates with lower uncertainty, such as BTLS, lead to smaller relative errors. Finally, the same pair of complex poles identified by BTLS in the *Linear Estimators Comparison* section is consistently recovered by the Newton-Gauss optimizer.

Nevertheless, when examining the poles obtained using Newton-Gauss with initial parameters from GTLS and BTLS (Tables 3.1 and 3.2), some unstable and highly damped poles with significant uncertainty level still appear, indicating that the model has not fully converged to the underlying system. We will therefore consider the final estimator, Levenberg-Marquardt, to achieve convergence to the true system model.

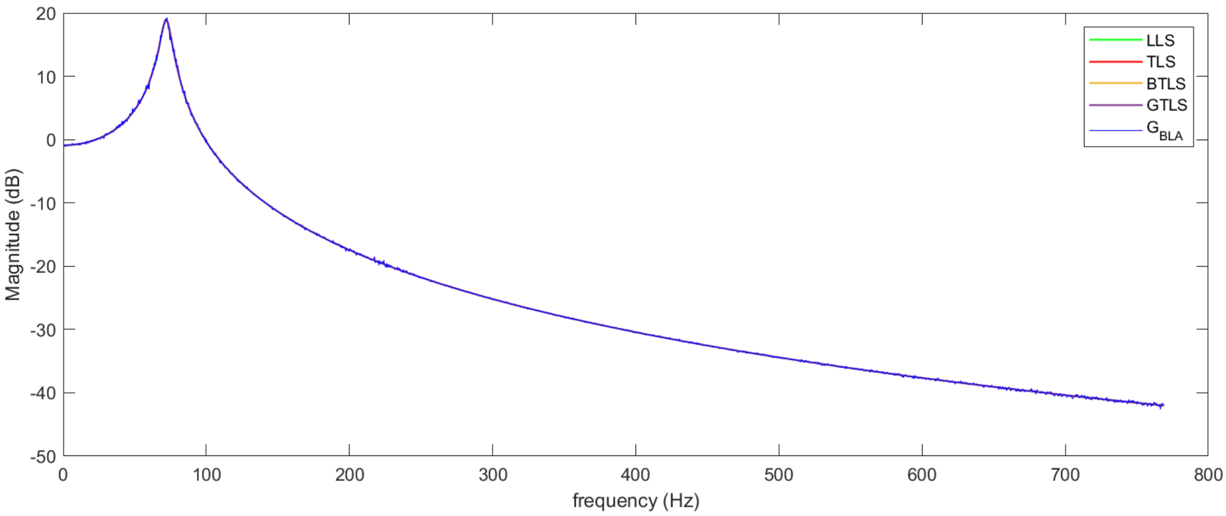


Figure 3.9: NG Estimator - TF Magnitude estimation - BIC

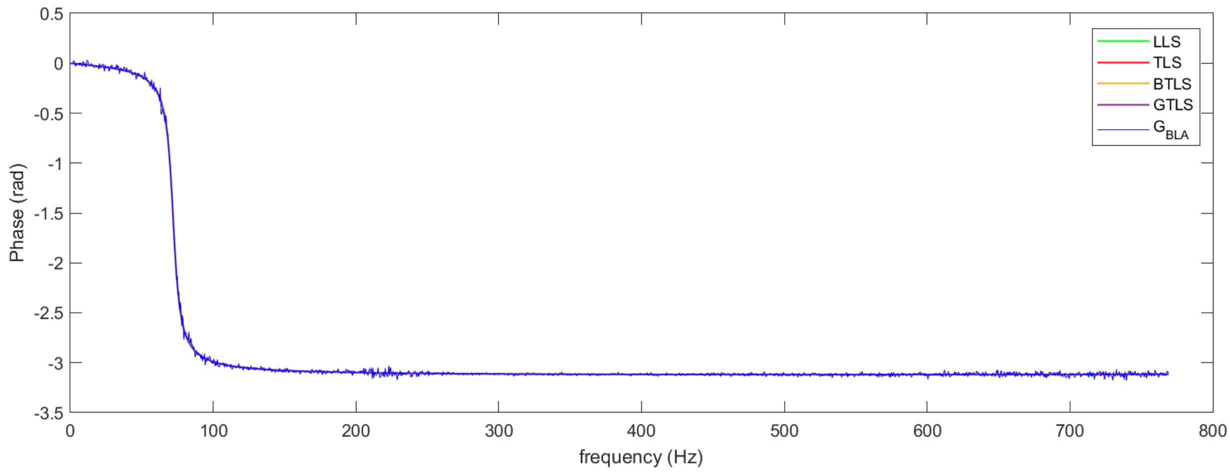


Figure 3.10: NG Estimator - TF phase estimation - BIC

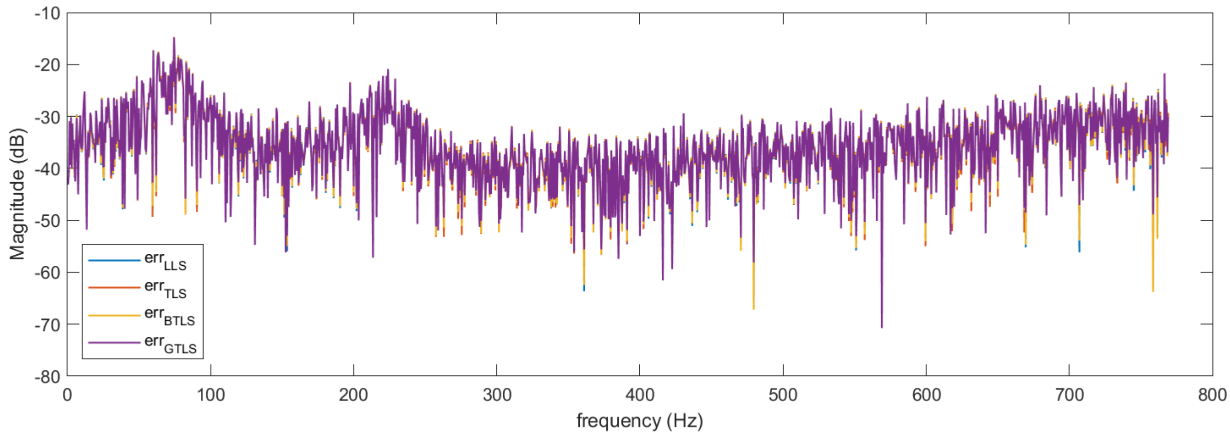


Figure 3.11: NG Estimator - Error - BIC

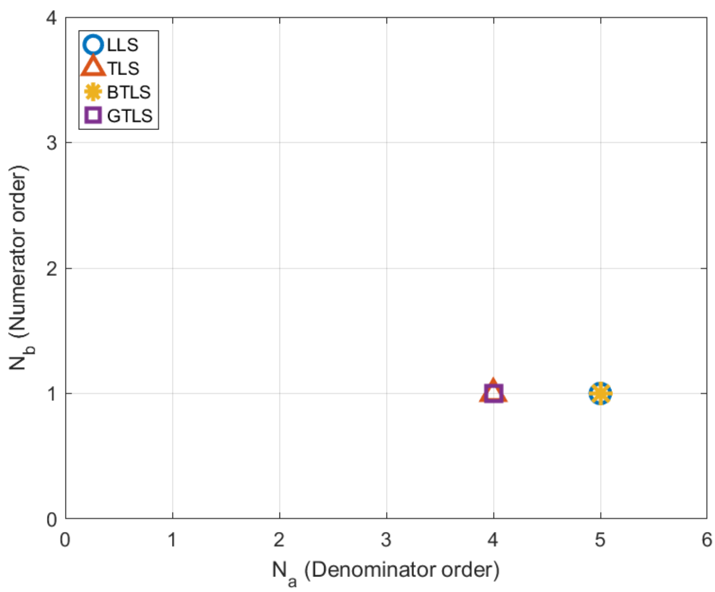
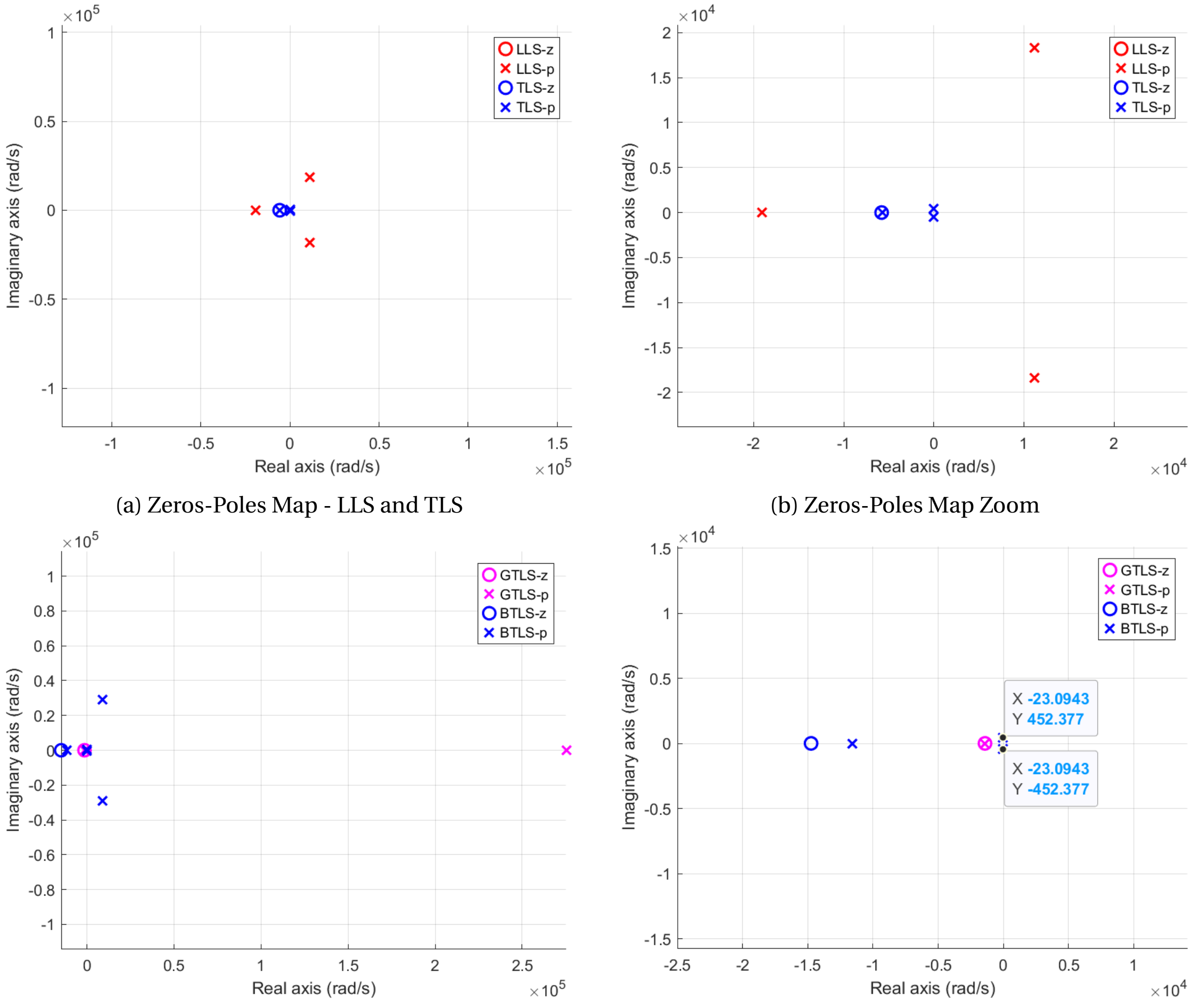


Figure 3.12: NG Estimator - Parameters space dimension - BIC



(a) Zeros-Poles Map - LLS and TLS

(b) Zeros-Poles Map Zoom

(c) Zeros-Poles Map - GTLS and BTLS

(d) Zeros-Poles Map Zoom

Figure 3.13: NG estimators – zeros–poles maps

Poles position	Standard deviation
275242.2	$1882.8 + j15.71$
-1390.8	$9.5 + j15.71$
$-23.3 + j452.5$	$0.2 + j3.1$
$-23.3 - j452.5$	$0.2 + j3.1$

Table 3.1: Poles uncertainty - Newton-Gauss with GTLS parameters as initial estimate

Poles position	Standard deviation
$8961.5 + j28943.1$	$4.86 + j15.71$
$8961.5 - j28943.1$	$4.86 + j15.71$
-11614.76	6.3
$-23.1 + j452.37$	$1.3e^{-2} + j0.25$
$-23.1 - j452.37$	$1.3e^{-2} + j0.25$

Table 3.2: Poles uncertainty - Newton-Gauss with BTLS parameters as initial estimate

Levenberg-Marquadt

This part will focus on the model estimate with GTLS and BTLS as first parameters computations. The Tables 3.3 and 3.4 shows a reduction of poles number converging to the (assumed) true poles of the system. However, *noisy* poles still appear. We can either try to increase the number of iterations in the LM algorithm or reduces the band of interest to filter out the unexcited dynamics in high frequencies. We decided to go on with the second choice which results to the same poles with extremely low uncertainty (10^{-13} order of magnitude) for both GTLS and BTLS initialization - Table 3.5. We can, therefore, assume that the true transfer function model contains a pair of complex poles and a pair of complex zeros - 3.6.

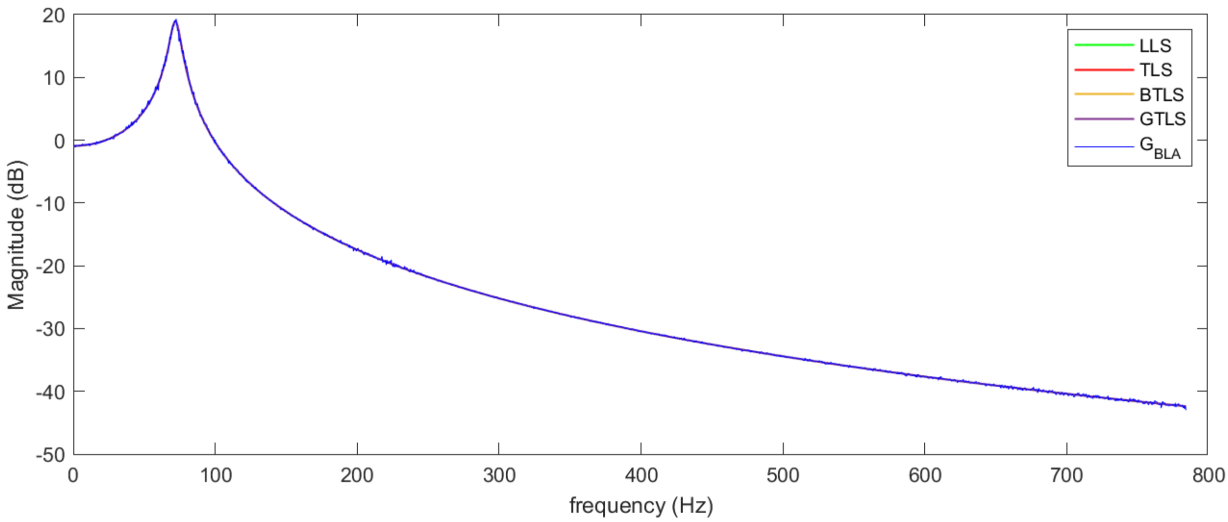


Figure 3.14: LM Estimators - TF Magnitude estimation - BIC

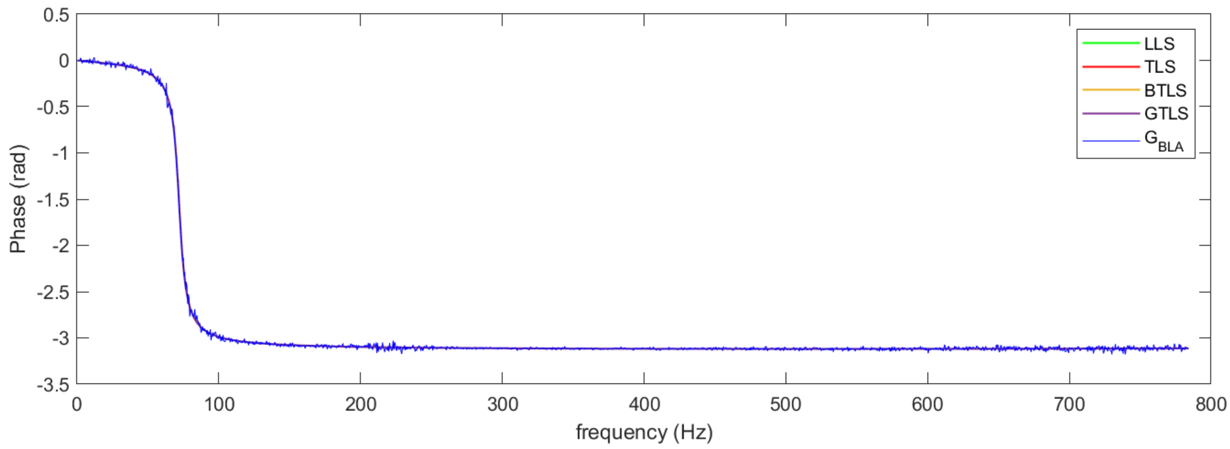


Figure 3.15: LM Estimators - TF phase estimation - BIC

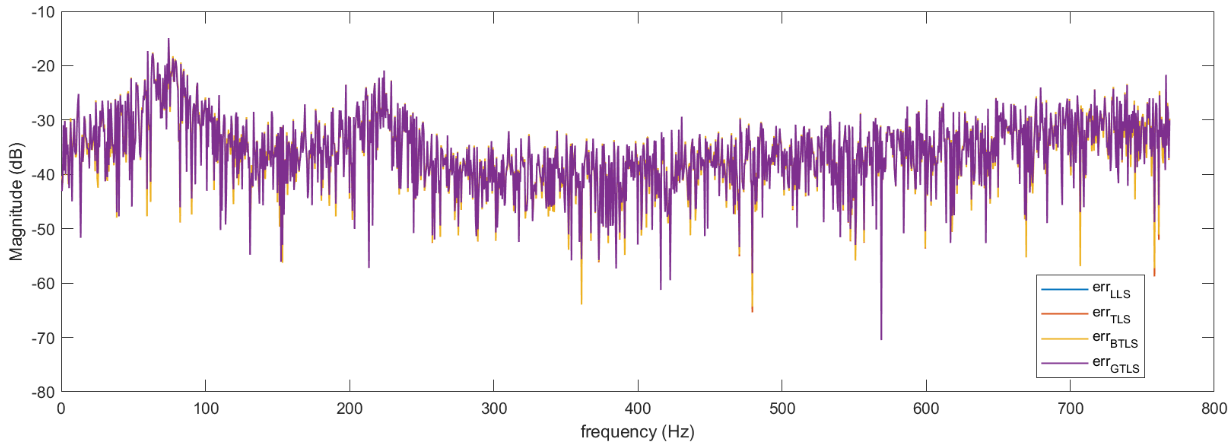


Figure 3.16: LM Estimators - Error - BIC

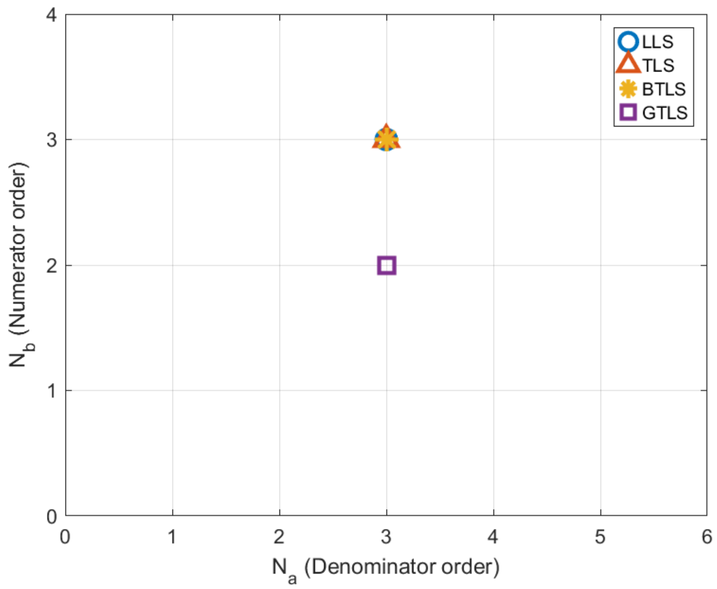
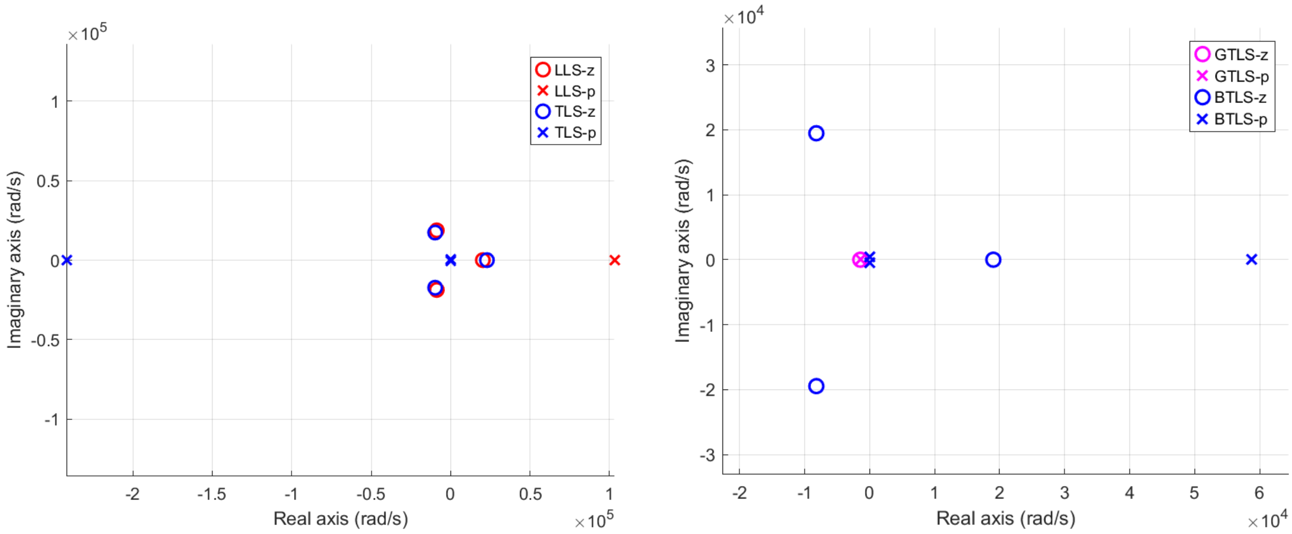


Figure 3.17: LM Estimator - Parameters space dimension - BIC



(a) Zeros-Poles Map - LLS and TLS

(b) Zeros-Poles Map - GTLS and BTLS

Figure 3.18: LM estimators – zeros–poles maps

Poles position	Standard deviation
-1437.1	12.79
$-23.3+j452.5$	$0.2+j4.02$
$-23.3-j452.5$	$0.2+j4.02$

Table 3.3: Poles uncertainty - Levenberg-Marquadt with GTLS parameters as initial estimate

Poles position	Standard deviation
58768.3	17.5
$-23.1+j452.37$	$6.6e^{-3}+j0.13$
$-23.1-j452.37$	$6.6e^{-3}+j0.13$

Table 3.4: Poles uncertainty - Levenberg-Marquadt with BTLS parameters as initial estimate

Restraining the band of interest up to 640Hz

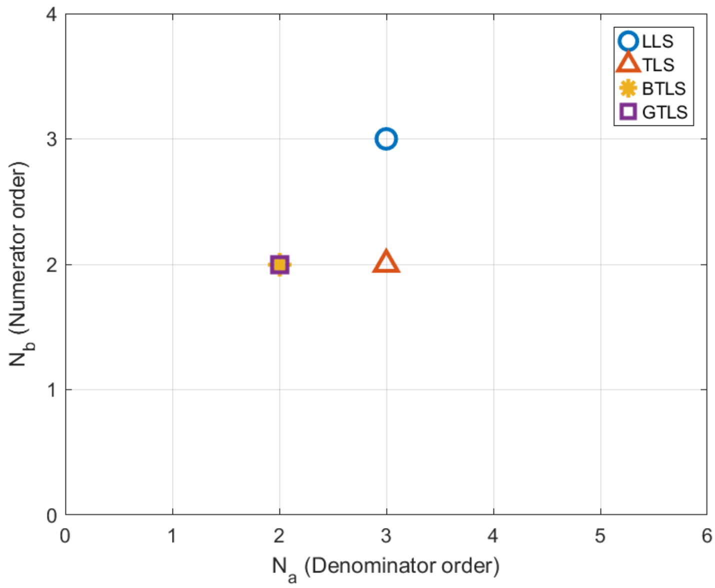
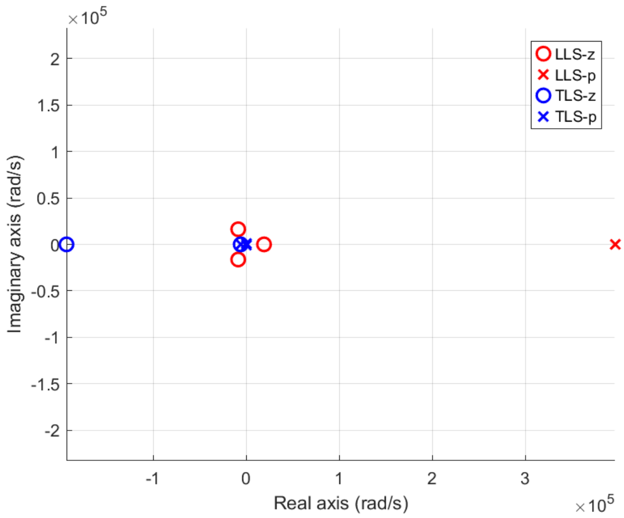
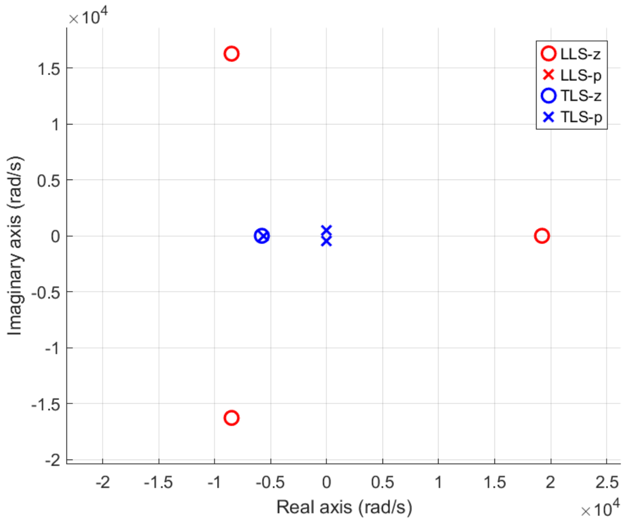


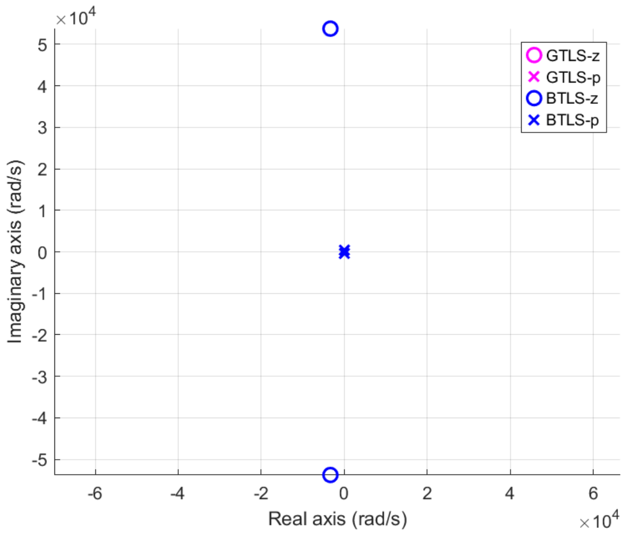
Figure 3.19: LM Estimator - Parameters space dimension - Restrained Bandwidth



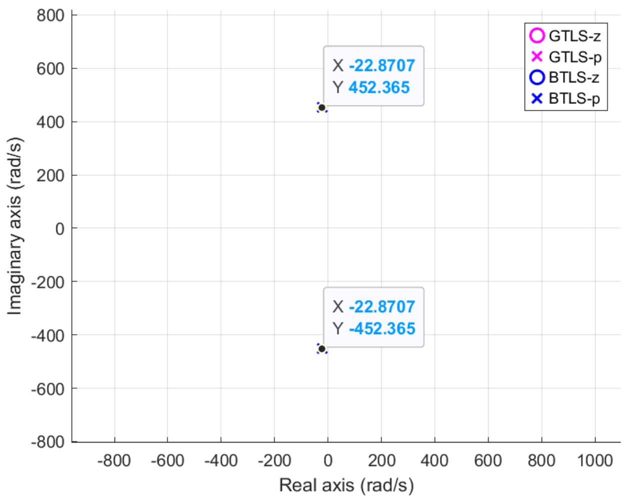
(a) Zeros-Poles Map - LLS and TLS



(b) Zeros-Poles Map Zoom



(c) Zeros-Poles Map - GTLS and BTLS



(d) Zeros-Poles Map Zoom

Figure 3.20: LM estimator – zeros–poles maps - Restrained Bandwidth

Poles position	Standard deviation
$-22.87 + j452.4$	$5.8e^{-14} + j9.5e^{-13}$
$-22.87 - j452.4$	$5.8e^{-14} + j9.5e^{-13}$

Table 3.5: Poles uncertainty - Levenberg-Marquadt with GTLS and BTLS parameters as initial estimate

Pole position	Zero position
$-22.87 + j452.4$	$-3.3e^{3} + j5.4e^{4}$
$-22.87 - j452.4$	$-3.3e^{3} - j5.4e^{4}$

Table 3.6: Zeros and Poles - Levenberg-Marquadt with GTLS and BTLS parameters as initial estimate

4 Conclusion

This report relates the progress of an unknow system achieved over more than nine weeks of measurements, researches and Matlab coding but, more importantly, nine weeks of reflection about the identification methodologies (parametric and non-parametric) seen during the lectures. Implementing these identification approaches reinforced a practical understanding of the roles, strengths, and limitations of both linear and nonlinear estimators. Moreover, working with real physical measurement devices allowed the modeling of an unknown system from scratch as it would be done in concrete real cases bridging the gap between theory and practice.